

Seguono b 1) e 2) :

$$3) \{A_n\} \in \mathcal{F} \quad A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots, \\ A_n \subseteq A_{n+1} \quad \forall n$$

$$\Rightarrow \mu\left(\bigcup_n A_n\right) = \lim_{n \rightarrow +\infty} \mu(A_n)$$

$$4) A_1 \in \mathcal{F}, \mu(A_1) < +\infty$$

$$\{A_n\} \in \mathcal{F} \quad A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots, \\ A_n \supseteq A_{n+1} \quad \forall n$$

$$\Rightarrow \mu\left(\bigcap_n A_n\right) = \lim_{n \rightarrow +\infty} \mu(A_n)$$

Quali sono gli insiemi di misura nulla?

• $(n=1)$

$$\rightarrow A = \{0\} \Rightarrow m(A) = 0$$

$$\{0\} = \bigcap_{n \in \mathbb{N}} \bigcap_{0 \leq y \leq 1} \left(-\frac{1}{n}, \frac{1}{n}\right) \Rightarrow$$

$$m(\{0\}) = \lim_{n \rightarrow +\infty} m\left(-\frac{1}{n}, \frac{1}{n}\right) = \lim_{n \rightarrow +\infty} \frac{2}{n} = 0$$

$$\rightarrow A = \{x_1, x_2, \dots, x_N\}$$

$$\rightarrow A = \{x_n\}_{n \in \mathbb{N}} \quad m(A) = \sum_n m(\{x_n\}) = 0$$

$$\rightarrow A = \mathbb{N}, \mathbb{Z}, \mathbb{Q}$$

$$\rightarrow A = \text{insieme di Cantor}$$

$$\rightarrow A = (0, 1) \setminus \mathbb{Q}$$

$$m(A) = ?$$

$$(0, 1) = ((0, 1) \cap \mathbb{Q}) \cup \overbrace{((0, 1) \cap (\mathbb{R} \setminus \mathbb{Q}))}^{\substack{\text{disgiunte} \\ A}}$$

$$\Rightarrow \underbrace{m(0, 1)}_1 = \underbrace{m((0, 1) \cap \mathbb{Q})}_0 + m(A)$$

$$\Rightarrow m(A) = 1$$

$$\rightarrow A = (0, +\infty)$$

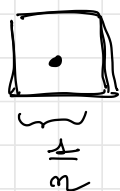
$$A = \bigcup_n \underbrace{(0, \frac{1}{n})}_{A_n}$$

$$A_n \subseteq A_{n+1}$$

$$m(A) = \lim_{n \rightarrow +\infty} \underbrace{m(A_n)}_{\frac{1}{n}} = +\infty$$

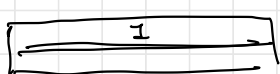
$$\boxed{n=2}$$

$$\rightarrow A = \{0\} \Rightarrow m(A) = 0$$



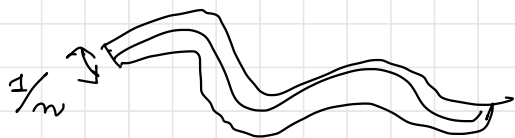
$$\bigcap_n A_n = \{0\}$$

$$\lim_{n \rightarrow +\infty} \underbrace{m(A_n)}_{\frac{1}{n^2}} = 0$$

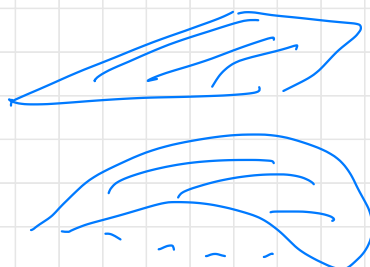
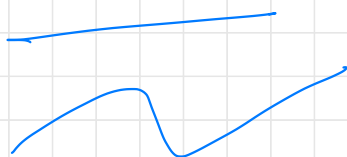


$$\bigcap_n A_n = \{S\}$$

$$m(S) = \lim_{n \rightarrow +\infty} \underbrace{m(A_n)}_{\frac{1}{n}} = 0$$



$$\boxed{n=3}$$



Df. Sia $f: \mathbb{R}^n \rightarrow \mathbb{R}$. Si dice che f è

MISURABILE secondo LEBESGUE se

$\forall A \subseteq \mathbb{R}$, $f^{-1}(A)$ è un misurabile secondo Lebesgue
aperto
" $\{x \in \mathbb{R}^n: f(x) \in A\}$

Op.

• è equivalente:

$\forall C \subseteq \mathbb{R}$, $f^{-1}(C)$ è un misurabile secondo Lebesgue
chiuso

• f continua $\Rightarrow$ misurabile

perché $\forall A$ aperto, $f^{-1}(A)$ aperto, quindi misurabile.

• $f: E$ misurabile $\subseteq \mathbb{R}^n \rightarrow \mathbb{R}$

f misurabile se

$\forall A$ aperto $\subseteq \mathbb{R}$ $\{x \in E: f(x) \in A\}$ misurabile
" $f^{-1}(A) \cap E$.

Integrale x cont Lebesgue

Pre $f: E \text{ misurabile} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, f misurabile

① $\text{SEMPLI} \geq 0$

② $f \geq 0$

③ f qualsiasi (tali che almeno uno tra $\int_E f^+$ e $\int_E f^-$ n  finito)

① $\mathcal{S}_1 = \{ \text{funzioni che assumono un n  finito di valori} \}$

$$s(x) = \begin{cases} \alpha_1 & x \in E_1 \\ \vdots & \vdots \\ \alpha_N & x \in E_N \end{cases} \quad \left. \begin{matrix} \text{2 a 2} \\ \text{disgiunti} \end{matrix} \right\}$$

$$s(x) = \sum_{i=1}^N \alpha_i \chi_{E_i}$$

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

$\uparrow$ funzione caratteristica di E .

$$s^{-1}(\text{dai } \alpha_i) = E_i \text{ misurabile.}$$

$$\int_E S \stackrel{\text{def}}{=} \sum_{i=1}^N \alpha_i m(E_i)$$

• convenzione $0 \cdot \infty = 0$

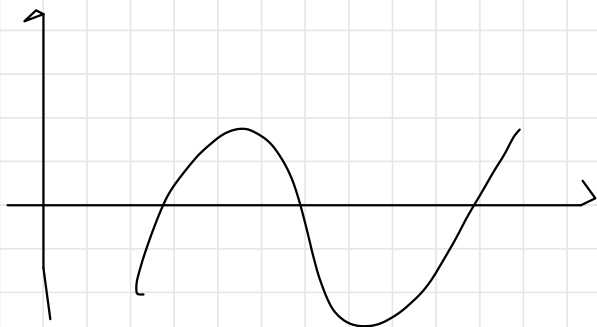
• $m \quad +\infty \quad -\infty$

② $f \geq 0$

$$\int_E f \stackrel{\text{def}}{=} \sup_{\substack{S \in \mathcal{S}_L \\ f \geq S \geq 0}} \int_E S$$

$$\left(= \inf_{\substack{S \in \mathcal{S}_L \\ S \geq f}} \int_E S \right)$$

③ f "qualsiasi"

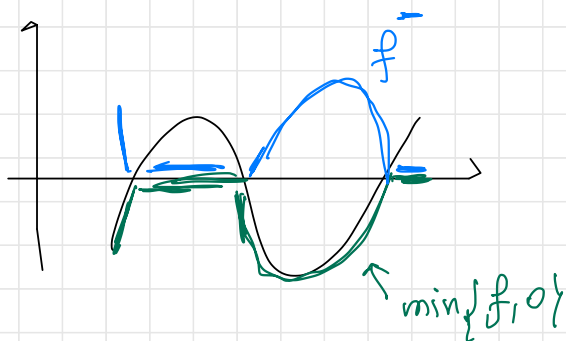
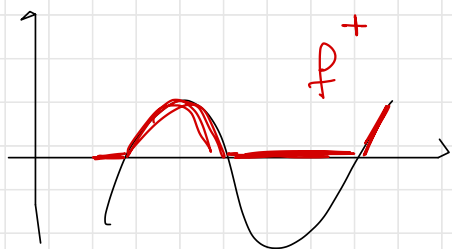


$$f = f^+ - f^-$$

$$f^+ = \max\{f, 0\}$$

$$f^- = -\min\{f, 0\}$$

$$|f| = f^+ + f^-$$



$$\int_E f \stackrel{\text{def}}{=} \int_E f^+ - \int_E f^-$$



a patto che almeno uno tra $\int_E f^+$ e $\int_E f^-$ sia finito.

Def. $f : E \rightarrow \mathbb{R}$ si dice
mis mis

INTEGRABILE secondo Lebesgue se

$\int_E f$ ha senso (ovvero almeno uno tra $\int_E f^+$ e $\int_E f^-$ è finito)

e $\int_E f \in \mathbb{R}$.

Dim.

f è integrabile secondo Lebesgue

$\iff f^+ \text{ e } f^- \text{ sono integrabili secondo Lebesgue}$

Siccome $|f| = f^+ + f^-$

se f integrabile secondo Lebesgue,

anche $|f|$ integrabile secondo Lebesgue

e viceversa.

Proprietà

1) linearità f, g integrabili, $\alpha, \beta \in \mathbb{R} \Rightarrow$
 $\alpha f + \beta g$ integrabile e $\int_E (\alpha f + \beta g) = \alpha \int_E f + \beta \int_E g$

2) monotonia f, g integrabili $f \leq g$ su E
 $\Rightarrow \int_E f \leq \int_E g$
 $f(x) \leq g(x) \quad \forall x \in E \cap \mathbb{N}$
ovvero $m(x) \neq 0$
 $f(x) \leq g(x) \quad \forall x \text{ q.o. } x \in E$

3) maggiorazione modulo

f è integrabile $\Rightarrow$

$$\left| \int_E f \right| \leq \int_E |f|$$

Conseguenza di 2):

$$-|f| \leq f \leq |f|$$

$\Downarrow$ per monotonia

$$-\int_E |f| \leq \int_E f \leq \int_E |f|$$

ovvero :

$$\left| \int_E f \right| \leq \int_E |f|$$

4) L'integrale "non vede" gli insiemi di misura nulla
 $\int_E f(x) = \int_E g(x)$ su $E \setminus N$ $m(N) = 0$,

allora f integrabile secondo Lebesgue $\Leftrightarrow$

g integrabile secondo Lebesgue

e in tal caso $\int_E f = \int_E g$.

Def.

Un predicato $P(x)$ vale *quasi ovunque su E*
per q.o. $x \in E$

se $P(x)$ vale $\forall x \in E \setminus N$ dove $m(N) = 0$

Esempio

• $f(x) \leq g(x)$ per q.o. $x \in E$

significa

$f(x) \leq g(x) \quad \forall x \in E \setminus N, \quad m(N) = 0$

• $f(x) = 0$ per q.o. $x \in E$

significa

$f(x) = 0 \quad \forall x \in E \setminus N, \quad m(N) = 0.$

$$\rightarrow f = 0 \quad \text{q.o.} \quad mE \Rightarrow \int_E f = 0$$

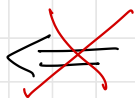
$$\rightarrow g = h \quad \text{q.o.} \quad mE \Rightarrow \int_E g = \int_E h$$

Confronto R-L

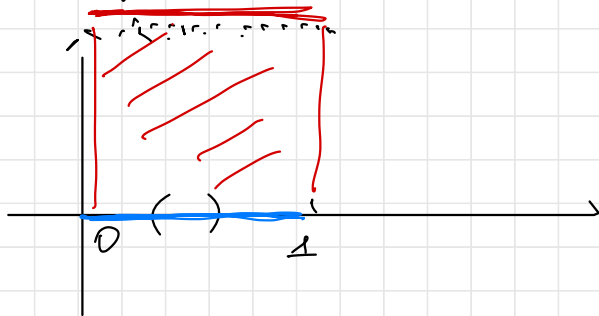
① f limitate & finite su limitati

② f illimitate e/o definite su illimitati

③ f R-integrabili $\Rightarrow f$ L-integrabili
(e in tal caso, i valori dei due integrali coincidono).



Controesempio



FUNZIONE DI
DIRICHLET

$$f(x) = \begin{cases} 1 & x \in \mathbb{Q} \\ 0 & x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

$$\begin{aligned} \int_0^1 f &= 1 \cdot m(\mathbb{Q} \cap (0, 1)) \\ &\quad + 0 \cdot m(\mathbb{R} \setminus \mathbb{Q} \cap (0, 1)) \\ &= 1 \cdot 0 + 0 \cdot 1 = 0 \end{aligned}$$

$$\mathcal{J}_L \supseteq \mathcal{J}_R$$

$\uparrow$ assumono # finito valori
 $\uparrow$ a scalo

$$f \geq 0$$

$$\sup_{s \in \mathcal{J}_R} \int_E f \quad \checkmark$$

$\alpha < s \leq \beta$

$$\sup_{s \in \mathcal{J}_L} \int_E s \quad (\leq)$$

$\alpha < s \leq \beta$

$$\inf_{s \in \mathcal{J}_L} \int_E s \quad \checkmark$$

$s \geq \beta$

$$\inf_{s \in \mathcal{J}_R} \int_E s$$

$s \geq \beta$



② f limitate ma definite su $\mathbb{R}$

Supp. f $\mathbb{R}$ -integrabile su $(-L, L) \quad \forall L > 0$.

f L -integrabile su $\mathbb{R}$

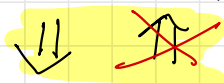


f L -integrabile su $\mathbb{R}$



f $\mathbb{R}$ -integrabile su $\mathbb{R}$
(in senso improprio)

(e in tal caso l'integrale di f
secondo Lebesgue e
secondo Riemann coincidono)



f $\mathbb{R}$ -integrabile su $\mathbb{R}$

Controesempio:

$$f(x) = \frac{\sin x}{x} \quad \text{su } (0, +\infty)$$

$\mathbb{R}$ -integrabile ma f non è $\mathbb{R}$ -integrabile.

Principali teoremi nell'integrazione cont. Lebesgue

Def. Sia $E \subseteq \mathbb{R}^n$ misurabile.

$L^1(E) \stackrel{\text{def}}{=} \left\{ f: E \rightarrow \mathbb{R} \text{ misurabili che sono } L\text{-integrabili su } E \right\}$
 $\uparrow$
fp. razionale

$$\forall f \in L^1(E), \quad \int_E f \in \mathbb{R}$$

$$f, g \in L^1(E), \quad \alpha, \beta \in \mathbb{R} \Rightarrow \alpha f + \beta g \in L^1(E)$$

$$\int_E \alpha f + \beta g = \alpha \int_E f + \beta \int_E g \in \mathbb{R}$$

Def. Data $f \in L^1(E)$ $\|f\|_1 := \int_E |f| \in \mathbb{R}$

$$1) \int_E |f| \geq 0 \text{ e } \int_E |f| = 0 \Rightarrow f = 0 \text{ q.o. m.E.}$$

$$2) \int_E |\alpha f| = |\alpha| \int_E |f|$$

$$3) \int_E |f+g| \leq \int_E |f| + \int_E |g|$$

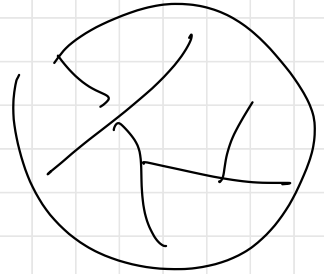
$$|f+g| \leq |f| + |g|$$

$$\int_E |f+g| \leq \int_E |f| + \int_E |g|$$

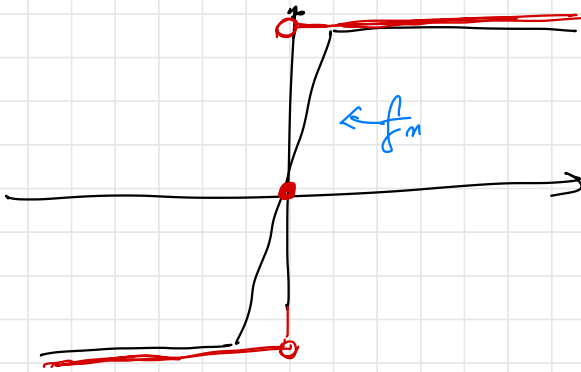
$$\text{Def } f \sim g \iff f(x) = g(x) \text{ q.o. su } E.$$

- $f \sim f$
- $f \sim g \iff g \sim f$
- $f \sim g, g \sim h \Rightarrow g \sim h$

$$[f] = \{g : g \sim f\}$$



Teorema $(L^1(E), \|\cdot\|_1)$ è uno spazio di Banach



Appartenenza a $L^1(E)$

$$f \in L^1(E) \iff \int_E |f| < +\infty$$

Convergenza in $L^1(E)$

$$f_n \in L^1(E), \quad f \in L^1(E)$$

$$f_n \longrightarrow f \text{ in } L^1(E) \quad \left(\lim_n f_n = f \text{ in } L^1(E) \right)$$

$\Updownarrow$ per def.

$$\lim_n \|f_n - f\|_1 = 0$$

ovvero

$$\left| \lim_n \int_E |f_n - f| = 0. \right|$$

Domanda:

$$f_n(x) \longrightarrow f(x) \quad \underbrace{\text{per q.o. } x \in E}_{\text{q.o. su } E}$$



$$\int_E |f_n - f| \longrightarrow 0$$

$$\lim_{n \rightarrow +\infty} \int_E |f_n - f| \stackrel{?}{=} \int_E \underbrace{\lim_n |f_n - f|}_0 = 0$$